

Deliverable 1 – Project Requirements

What is a Web Server?

A **web server** is software that handles requests from clients (usually web browsers) and delivers web content such as HTML pages, images, and files over HTTP/HTTPS. Example: Apache HTTP Server.

Web Server Applications

1. Apache HTTP Server

- **Definition:** Open-source web server widely used for hosting websites.
- **Website:** <https://httpd.apache.org/>
- **Operating Systems:** Linux, Windows, macOS
- **Latest Version:** 2.4.x

2. Nginx

- **Definition:** High-performance web server known for handling many connections efficiently.
- **Website:** <https://nginx.org/>
- **Operating Systems:** Linux, Windows, macOS
- **Latest Version:** 1.25.x

3. Microsoft IIS

- **Definition:** Web server developed by Microsoft for Windows environments.
 - **Website:** <https://www.iis.net/>
 - **Operating Systems:** Windows only
 - **Latest Version:** IIS 10
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What is Virtualization?

Virtualization is the process of creating a virtual version of a computer system, allowing multiple operating systems to run on a single physical machine.

What is VirtualBox?

VirtualBox is a free and open-source virtualization software that allows users to create and run virtual machines on their computer.

What is a Virtual Machine?

A **virtual machine (VM)** is a software-based computer that runs an operating system just like a physical computer.

Host Machine vs Guest Machine

- **Host Machine:** The physical computer running virtualization software.
 - **Guest Machine:** The virtual system running inside the host machine.
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What is Debian?

Debian is a free and open-source Linux operating system known for its stability and security.

What is a Firewall?

A **firewall** is a security system that monitors and controls incoming and outgoing network traffic based on rules.

What is SSH?

SSH (Secure Shell) is a protocol used to securely connect to remote computers over a network.

What is an IP Address?

An **IP address** is a unique identifier assigned to a device on a network.

Example: 192.168.1.1

What is a Network Mask?

A **network mask (subnet mask)** defines which portion of an IP address represents the network and which part represents the host.

Example: 255.255.255.0

What is a Port?

A **port** is a communication endpoint used by applications to send and receive data.

Examples:

- Port 80 → HTTP
 - Port 443 → HTTPS
 - Port 22 → SSH
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What is Port Forwarding?

Port forwarding is a technique that directs external traffic to a specific device or service within a private network.

What is Localhost?

Localhost refers to the local computer you are currently using.

What does 127.0.0.1 represent?

127.0.0.1 is the loopback IP address, which always refers to your own computer (localhost).

What is Git?

Git is a version control system used to track changes in code and collaborate with others.

What is GitHub?

GitHub is an online platform that hosts Git repositories and allows developers to collaborate, share, and manage code.

Concepts I Did Not Understand

1. Port Forwarding

At first, it was unclear how external traffic reaches internal devices. After research, I learned that routers map external ports to internal IP addresses.

2. Network Mask

Initially confusing, but I learned it separates network and host portions of an IP address.

3. Virtualization

I didn't fully understand how multiple systems run on one machine. Research showed virtualization uses software to simulate hardware environments.
